

Portfolio Project 2:

Manufacturing Defects Analysis – Pareto Approach

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Manufacturing Defect Analysis – Electronic Board Production

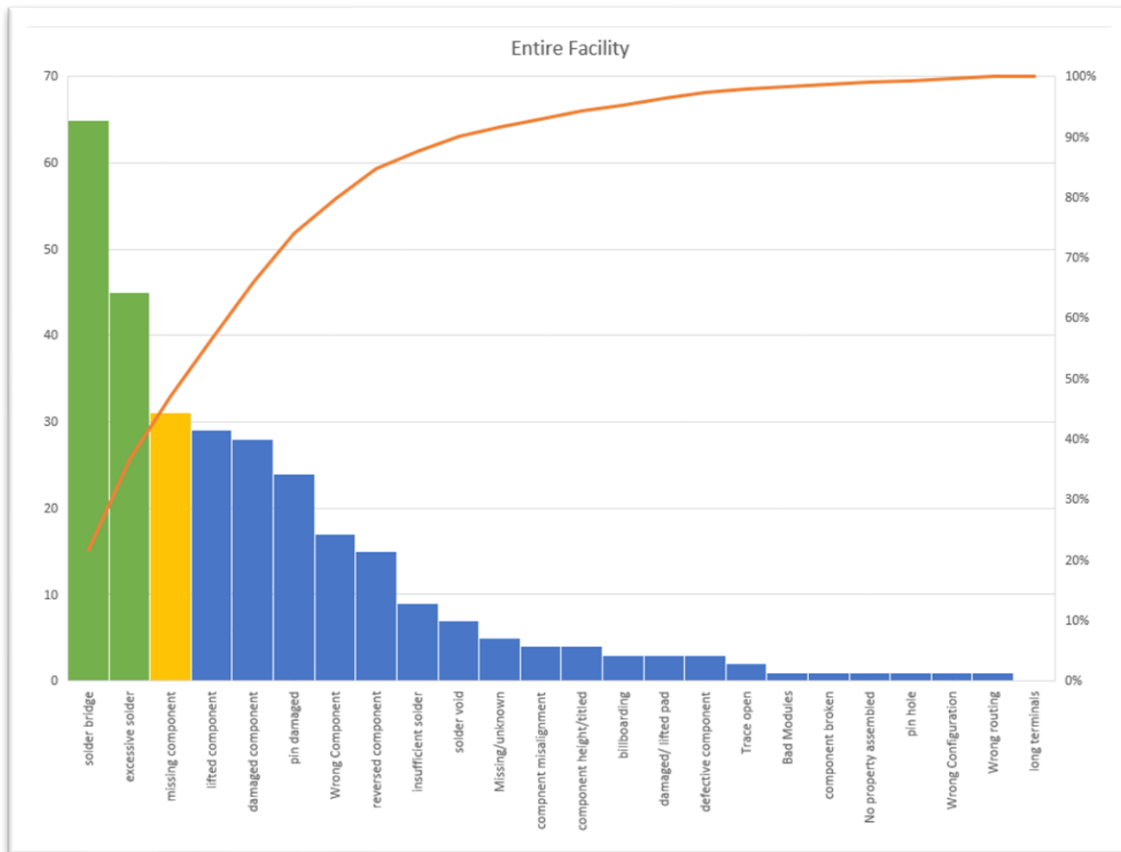
Problem Statement

The Manual Finish production area has experienced a rising number of welding defects that fail to meet IPC-A-610E standards, including solder bridges, insufficient solder, and lifted components. These defects compromise product reliability and assembly performance. Since post-production corrections require excessive labor hours and skilled technicians, improving defect prevention during the welding process is the most viable solution. A data-driven analysis is necessary to identify the root causes of defects, improve process controls, and optimize the welding workflow. The goal is to achieve a 20% reduction in defects and a 20% increase in production capacity without sacrificing quality.

Significant Causes of Defects (Pareto Analyses 1-4)

Analysis Approach: A Pareto analysis was conducted to identify the most frequent across specific models and overall defects. The results show that three types of defects are responsible for approximately percent of all welding-related issues.

Pareto Narratives and Analysis: Entire Facility

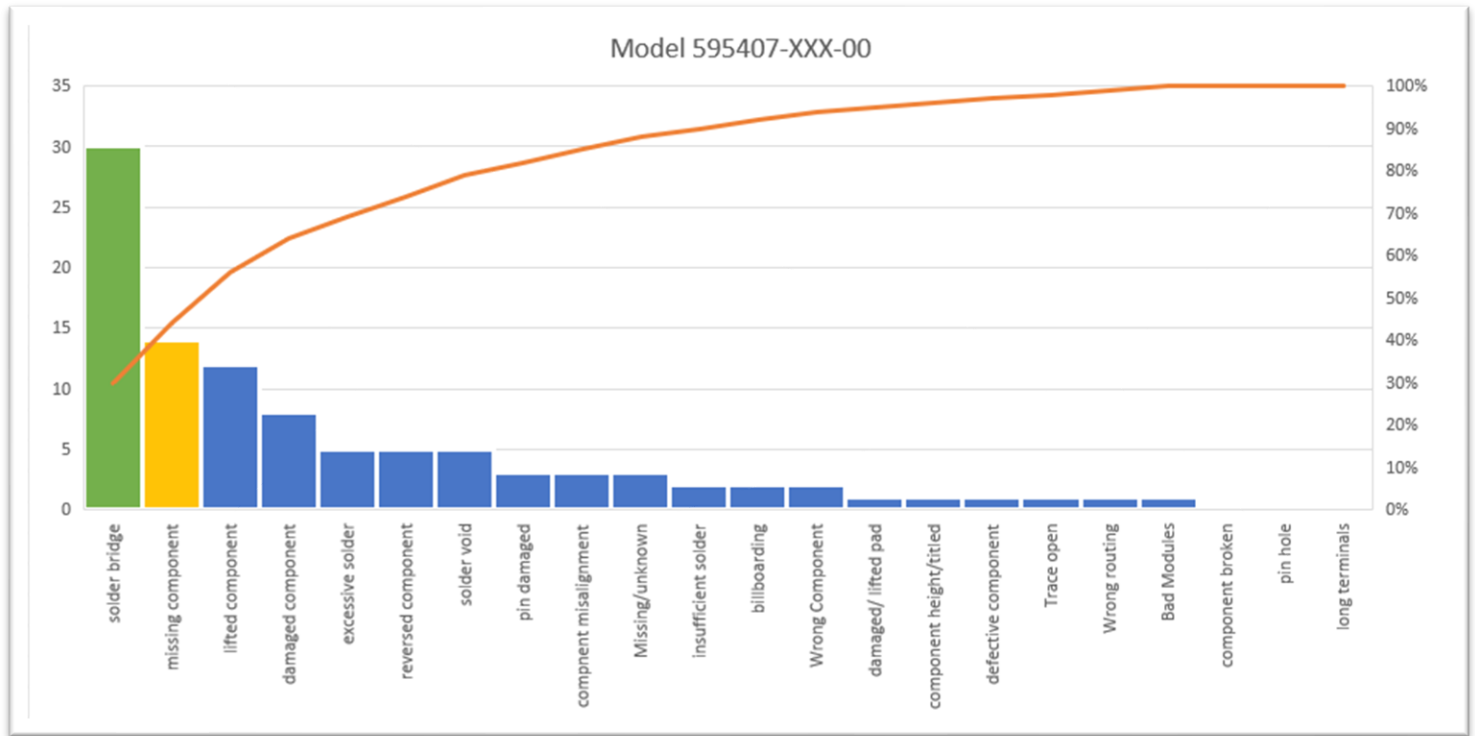


The three most frequent defects in the facility are:

- **Solder Bridge** (65 defects): Unwanted connections.
- **Excessive Solder** (45 defects): Too much solder causing issues.
- **Missing Component** (31 defects): Components not where they should be.

These account for 46% of all defects. Prioritizing these will significantly reduce overall defects.

Pareto Narratives and Analysis: Model 595407-XXX-00

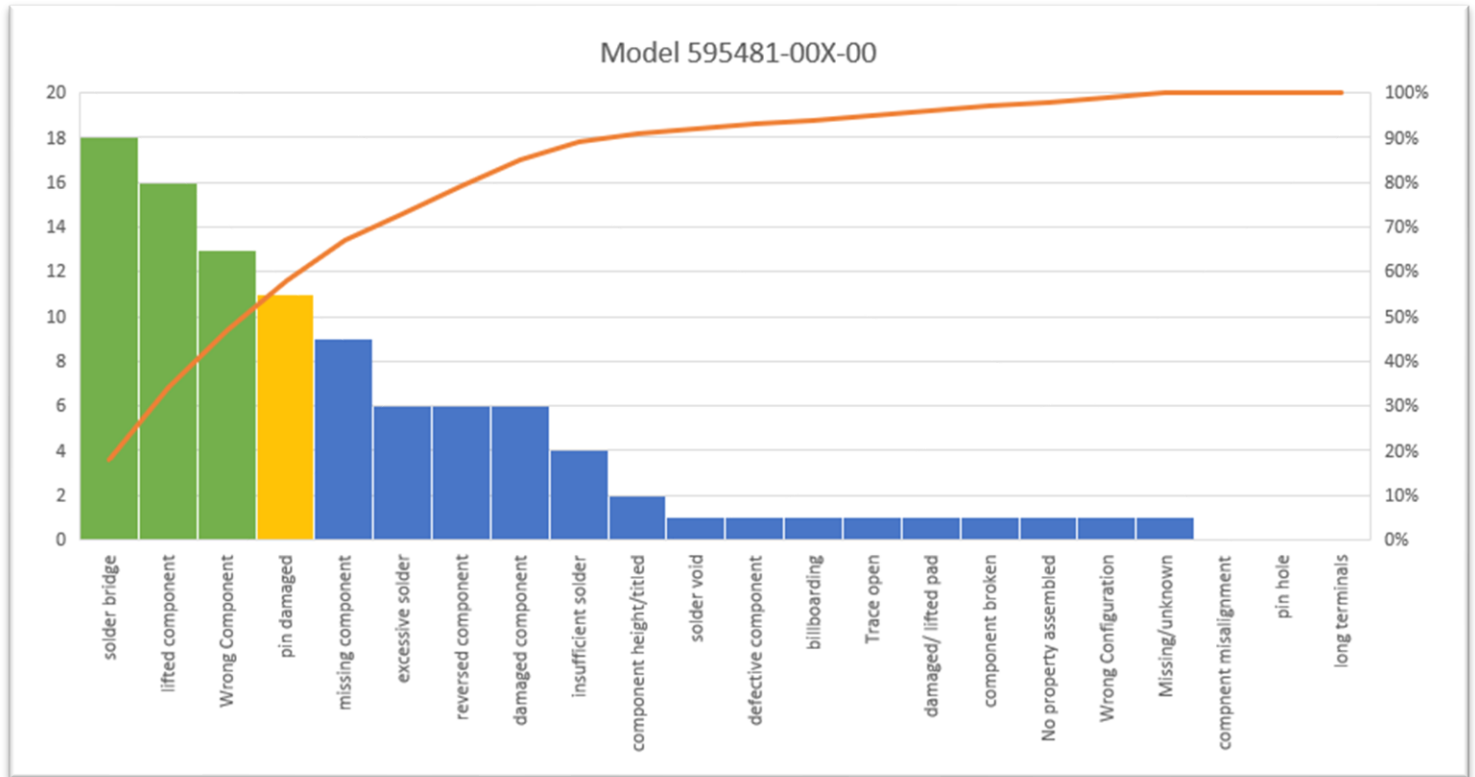


Top defects:

- **Solder Bridge** (30 defects): Leading issue.
- **Missing Component** (14 defects): Parts missing.
- **Lifted Component** (12 defects): Improper placement.

These make up 68% of defects for this model, indicating a need for process tightening.

Pareto Narratives and Analysis: Model 595481-00X-00

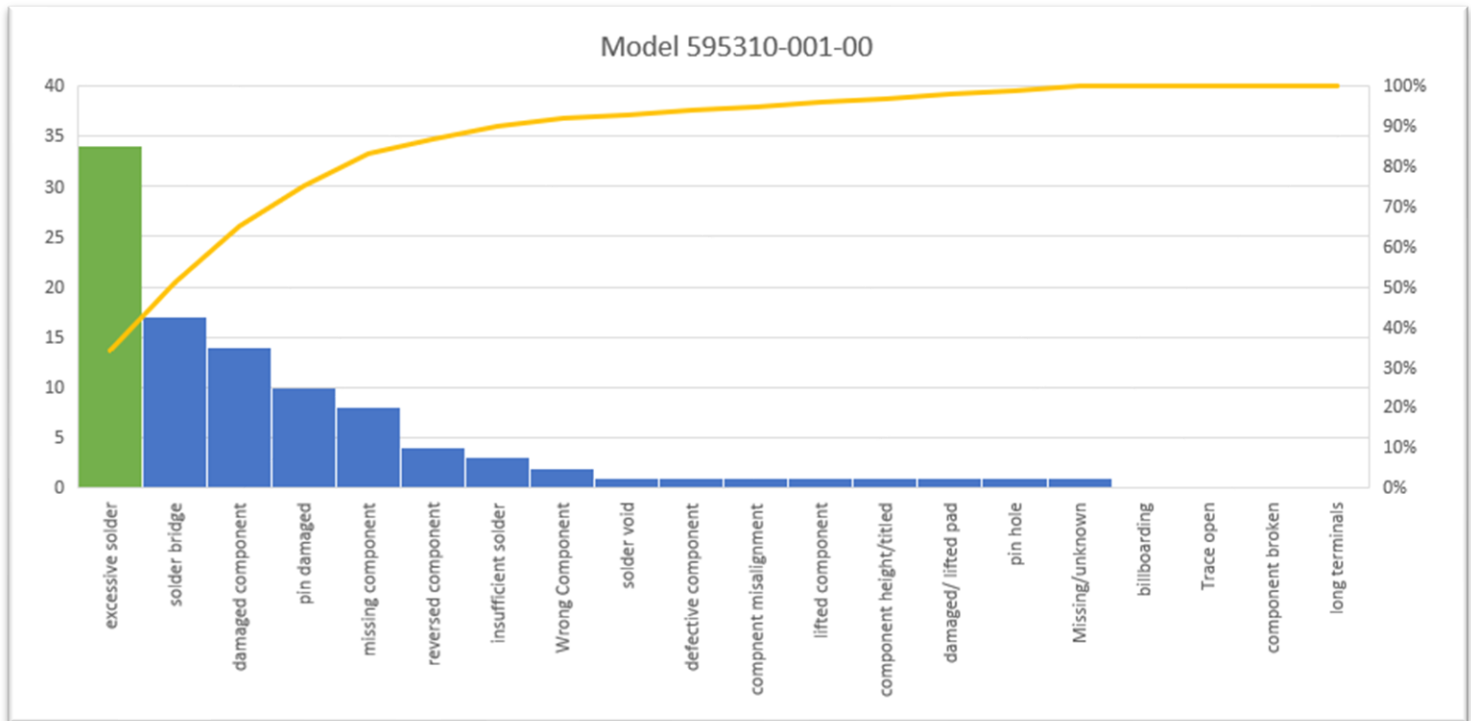


Top defects:

- **Solder Bridge** (18 defects): Major problem.
- **Lifted Component** (16 defects): Components not secured.
- **Wrong Component** (13 defects): Incorrect parts used.

These account for 56% of defects, suggesting improvements in material handling and accuracy.

Pareto Narratives and Analysis: Model 595310-001-00



Top defects:

- **Excessive Solder** (34 defects): Main issue.
- **Solder Bridge** (17 defects): Frequent problem.
- **Damaged Component** (14 defects): Mishandled parts.

These contribute to 74% of defects, highlighting the need for better process discipline and handling.

Next Steps for Resolution

Process Standardization

- **Solder Application Guidelines:** Clear rules to prevent excessive solder and bridges.
- **Component Placement Protocols:** Assigned placements for components.

Equipment Optimization

- **Machine Maintenance:** Regular maintenance required.
- **Wave Soldering Adjustments:** Fine-tuning necessary for precision.

Workforce Training

- **Operator Training:** Workshops on perfect soldering and placement.
- **Defect Feedback Loop:** Real-time alerts for early defect detection.

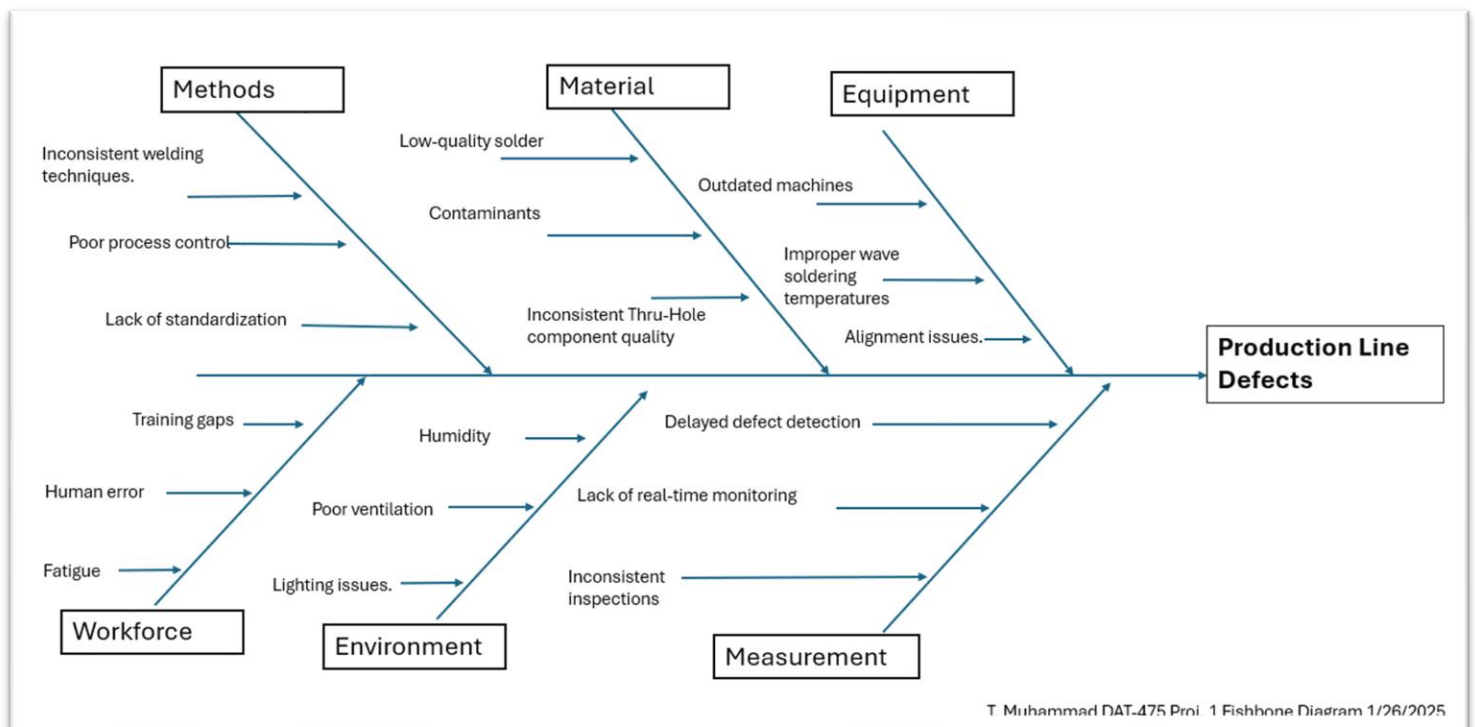
Quality Control Enhancements

- **Real-Time Monitoring:** Early defect detection technology.
- **Root Cause Tracking:** Keeping records of recurring defects for targeted actions.

Root Cause Analysis (Fishbone Diagram)

A root cause analysis was conducted to determine the underlying reasons for these defects. The fishbone diagram categorizes the causes into six main areas.

Root Causes of Defects



Production Line Comparison

A review of individual production lines revealed that defects are not evenly distributed across all lines.

- Line A has the highest rate of solder bridges, suggesting an issue with alignment or machine settings.
- Line B experiences frequent insufficient solder defects, likely due to inconsistencies in wave soldering.
- Line C shows a mix of lifted components and solder bridges, indicating handling or process issues.

Each production line may require targeted improvements based on its specific defect trends.

Conclusion and Next Steps

Key Findings

- The three most common defects, insufficient solder, solder bridges, and lifted components, account for 82% of the problems.
- The root causes include process inconsistencies, equipment malfunctions, and material issues.
- Different production lines exhibit different defect patterns, meaning a one-size-fits-all solution will not be effective.

Recommended Actions

- Standardize Processes – Implement strict soldering guidelines to reduce inconsistencies.
- Upgrade Equipment – Perform regular maintenance and calibration of soldering machines.
- Improve Training – Conduct hands-on soldering workshops to reduce human errors.
- Enhance Quality Monitoring – Introduce real-time defect detection to catch issues early.

By implementing these changes, the company can reduce defects, lower rework costs, and increase production efficiency.

Final Thoughts

Defect rates will not improve on their own. This analysis identifies the main issues, explains why they occur, and outlines a plan to address them. With the right improvements, the company can increase efficiency, reduce defects, and meet production demands.

Now is the time to put the plan into action and improve the production process.

References

American Society for Quality. (n.d.). *Plan-do-check-act cycle*. Retrieved from <https://asq.org/quality-resources/pdca-cycle>

Microsoft. (n.d.). *Create a Pareto chart*. Retrieved from <https://support.microsoft.com/en-us/office/create-a-pareto-chart-a1512496-6dba-4743-9ab1-df5012972856>